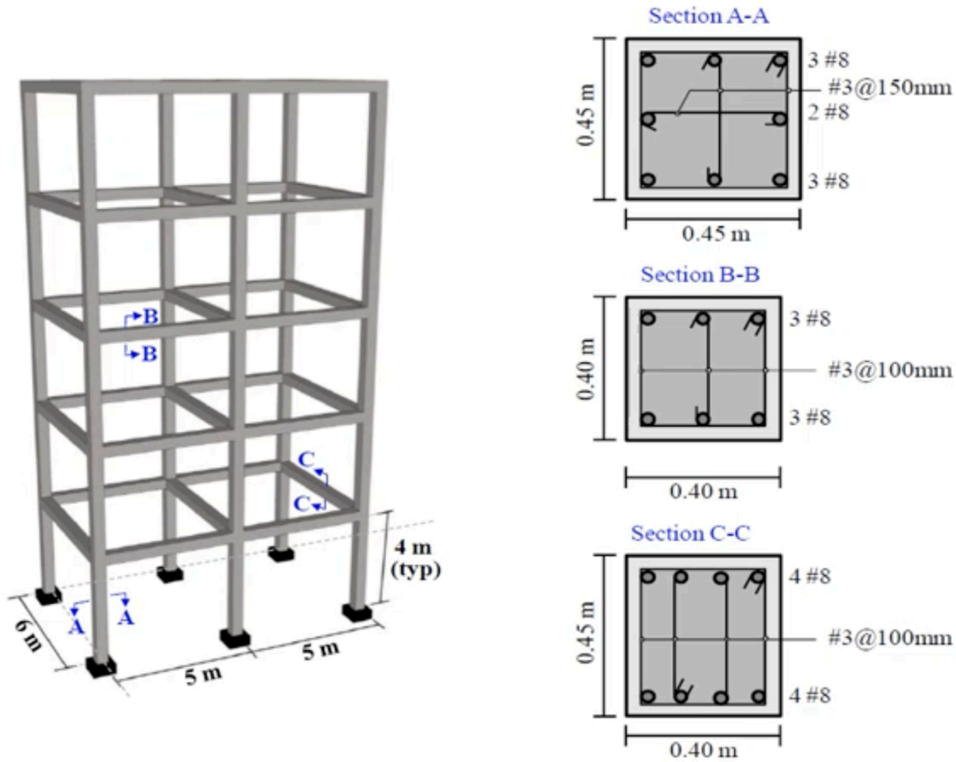
Session 1 Assignment (Full Frame Analysis**)**

Linear/Nonlinear Analysis of a 3D Frame

The geometrical properties of a building frame structure are shown in Figure 1.



For All Floor,
Dead Load = 6.5 kN/m^2 Live Load = 2.5 kN/m^2

Code: https://github.com/nirajkyadav/Homeworks/blob/main/Tutorial/Session_1_Assignment.py

Observations & Learnings:

Fundamental Periods: Mode 1 is a pure translational mode in the Z-direction, exciting 81.14% of the structure's dynamic mass while Mode 2 is a pure translational mode in the X-direction, exciting 82.14% of the mass. This directly correlates with the frame geometry, as the structure is more flexible in the Z-direction because it consists of only a single bay (6.0 m width), whereas the two-bay configuration in the X-direction (10.0 m total width) provides greater lateral stiffness.
Mode 3 is a torsional mode, activating 82.73% of the rotational mass about the vertical axis (RMY). And, Modes 4, 5, and 6 represent the higher-order flexure and torsion.

In seismic design codes (such as Nepal National Building Code 2020), an analysis must capture at least 90% of the structure's mass. The cumulative mass participation ratios by Mode 5, we successfully capture 92.94% in the X-direction and 92.32% in the Z-direction.

This also shows that a 2D analysis would have completely failed to capture the torsional behavior observed in Mode 3 and Mode 6.

Static Equilibrium Verification: The gravity analysis yields a total vertical reaction of 3997.68 kN across the six fixed base nodes. This total reaction matches the global sum of the slab dead/live loads combined with the self-weight of the concrete columns and beams, verifying equilibrium in the finite element model.

Center of mass: The computed center of mass of the structure is located at (5.0 m, 11.86 m, 3.0 m). The structure is symmetric in plan, so the center of mass is positioned at the geometric center in the horizontal directions. In the vertical direction, the center of mass is located at 11.86 m, which is slightly above the geometric mid-height of the structure (10 m). This occurs because no mass is assigned to the fixed base nodes. Consequently, only the superstructure contributes to the mass distribution, shifting the vertical center of mass upward.

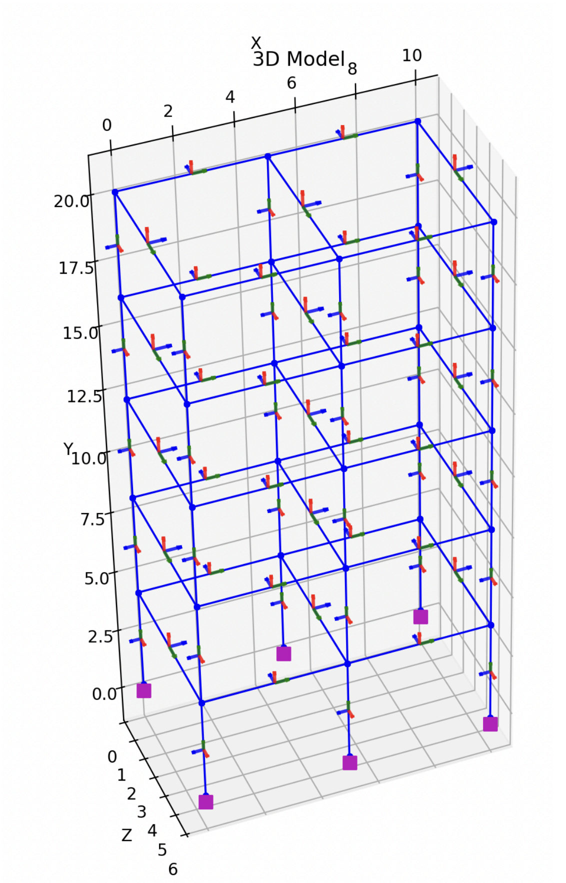
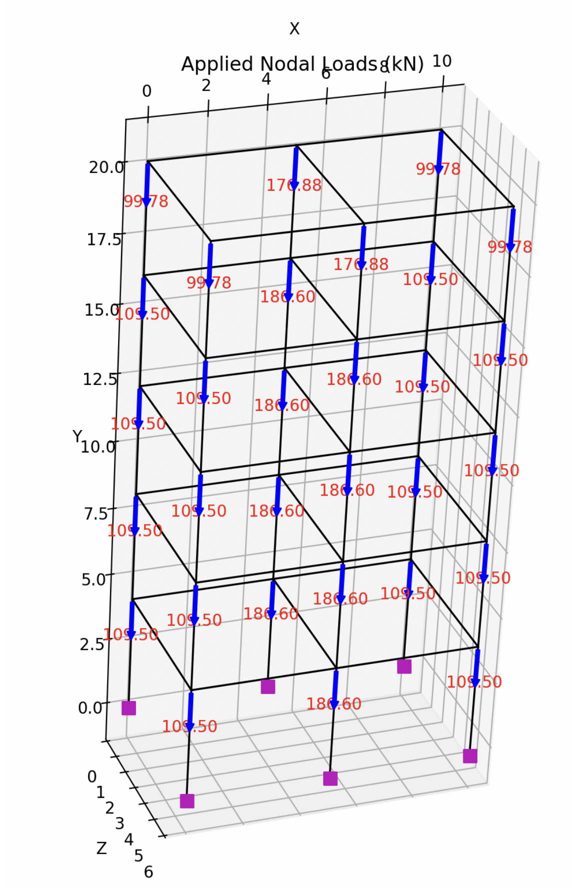
Time Periods	**Mode**
T_1 = 1.09643s	lateral translation in the Z-direction
T_2 = 0.984784s	lateral translation in the X-direction
T_3 = 0.863444s	torsional twist about the vertical Y-axis
T_4 = 0.338194s	
T_5 = 0.307721s	
T_6 = 0.274183s	

Total Vertical Reaction = 3997.68 kN

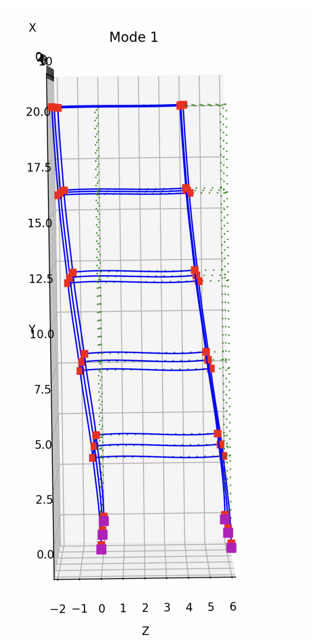
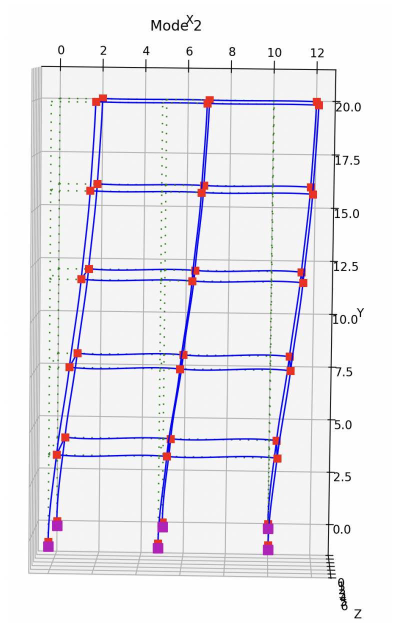
MODAL PARTICIPATION MASS RATIOS (%): The modal participation mass ratios (%) for each mode.

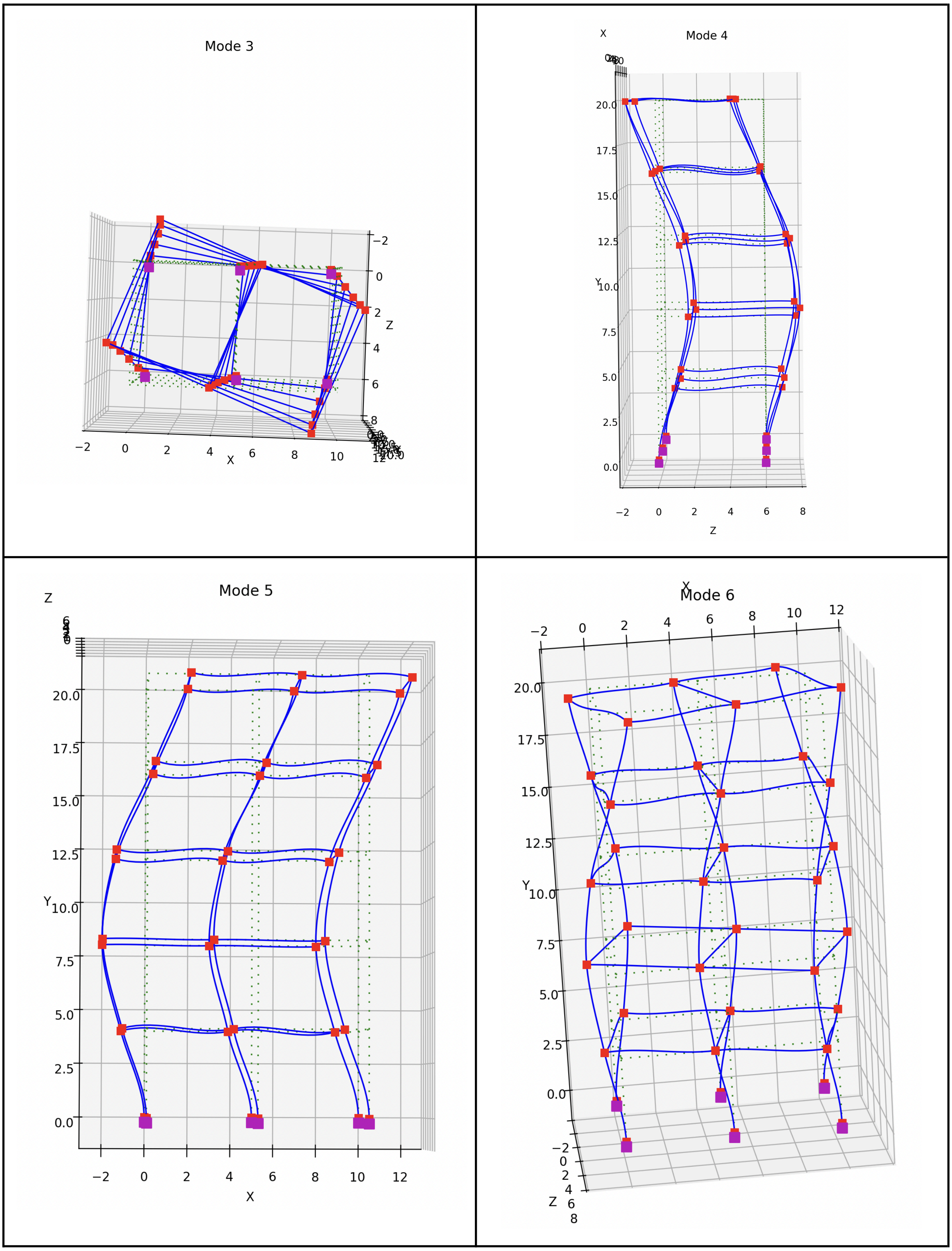
MODE	**MX**	**MY**	**MZ**	**RMX**	**RMY**	**RMZ**
1	0	0	81.1399	14.6469	0	0
2	82.1435	0	0	0	0	12.4109
3	0	0	0	0	82.733	0
4	0	0	11.1822	47.6775	0	0
5	10.7988	0	0	0	0	44.7778
6	0	0	0	0	10.3887	0

Diagrams:

3D Model	Applied Nodal Loads
	

Mode Shapes:


Mode 3
X
Z
Mode 4
X
Y
Z
Mode 5
Z
Y
X
Mode 6
X
Y
Z

Session 2 Assignment (Moment Curvature Analysis**)**

Assignment

- Moment-curvature analysis
  - Monotonic loading
    - Develop moment-curvature diagram
      - Displacement control integrator
      - Load control integrator
  - Cyclic loading
    - Develop moment-curvature diagram
      - Try different steel and concrete materials
      - Try different nonlinear analysis algorithms
      - Try different convergence tests and tolerances

Observations & Learnings:

1) Monotonic Loading:

The displacement-controlled analyses (Figures 2, 3, 4, and 5) successfully trace the complete moment-curvature response. The curves pass smoothly through the peak load, capture the subsequent strength degradation caused by concrete cracking and softening, and continue along the descending branch to curvatures exceeding 0.0012 rad/in.

In contrast, in load-controlled analyses, when Concrete02 (which incorporates concrete softening) is used, it terminates at the onset of strength degradation (Figures 7 and 9). These analyses are unable to proceed beyond the peak because the applied load exceeds the section's reduced load-carrying capacity. Consequently, the descending branch of the response cannot be captured. This limitation is not observed with Concrete01 (Figures 6 and 8), as this material model does not exhibit post-peak softening, allowing the load-controlled analysis to converge throughout the loading history.

These results show that load-controlled integrators are only suitable for monotonic hardening behavior. They are incapable of capturing post-peak material degradation, crushing, or softening because, on the descending branch, a single load level may correspond to multiple deformation states which a load-controlled algorithm cannot determine.

This assignment clearly illustrates why displacement (curvature) control is the preferred approach for generating moment-curvature diagrams. By using curvature increments rather than load increments, the solver can continuously track the structural response through both the ascending and descending branches, accurately capturing hardening, peak strength, and post-peak softening behavior.

Displacement control integrator

Code: https://github.com/nirajkyadav/Homeworks/blob/main/Tutorial/Session_2_Monotonic.py

Locking the axial DOF forces the axial strain at the centroid to be exactly zero. As a result, the cross-section is prevented from undergoing its natural axial expansion during bending. To satisfy this constraint, OpenSees develops a significant artificial compressive axial force within the section. This compression increases the load carrying capacity of the concrete fibers which can be seen in Figure 1.

This doesn't match the intended curve. Upon releasing the axial DOF allows the section to deform freely in the axial direction. This ensures that the resultant axial force across the fiber cross-section remains effectively zero. Under this condition, the computed response matches the accurate moment curvature analysis curve which can be seen in Figure 2.

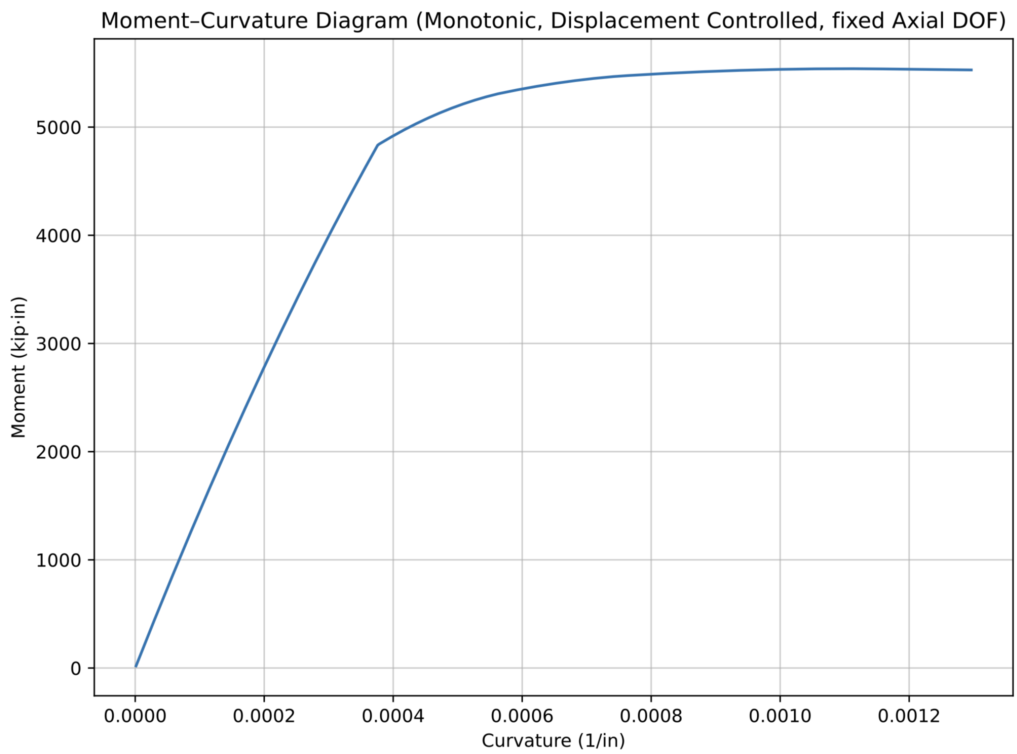


Figure 1. Moment Curvature Diagram (fixed Axial DOF), using Steel 01 & Concrete 01

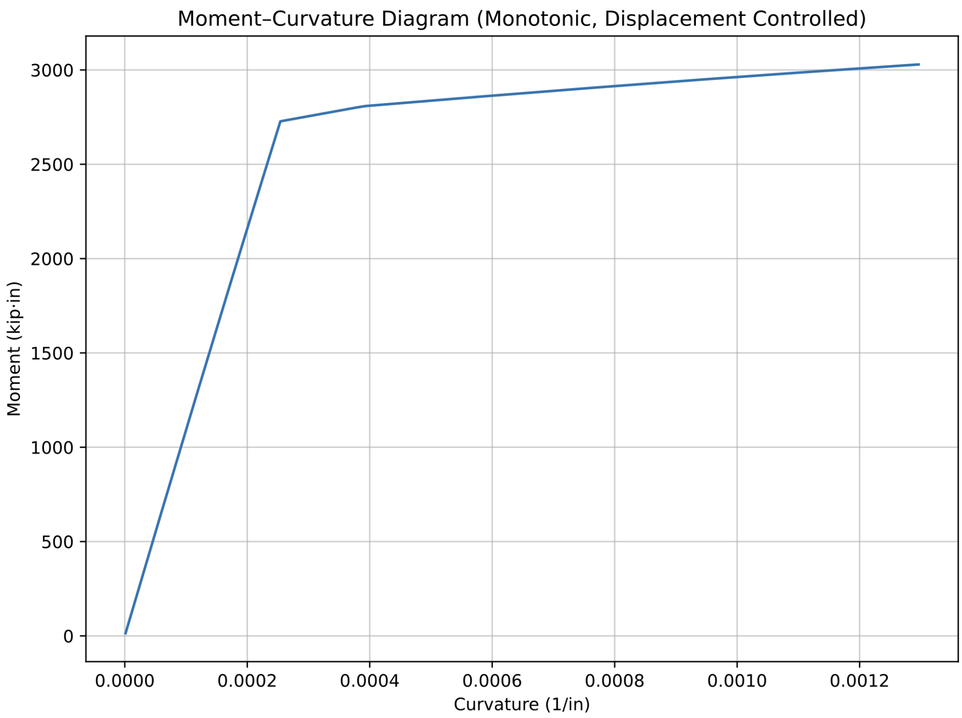


Figure 2. Moment Curvature Diagram (released Axial DOF), using Steel 01 & Concrete 01

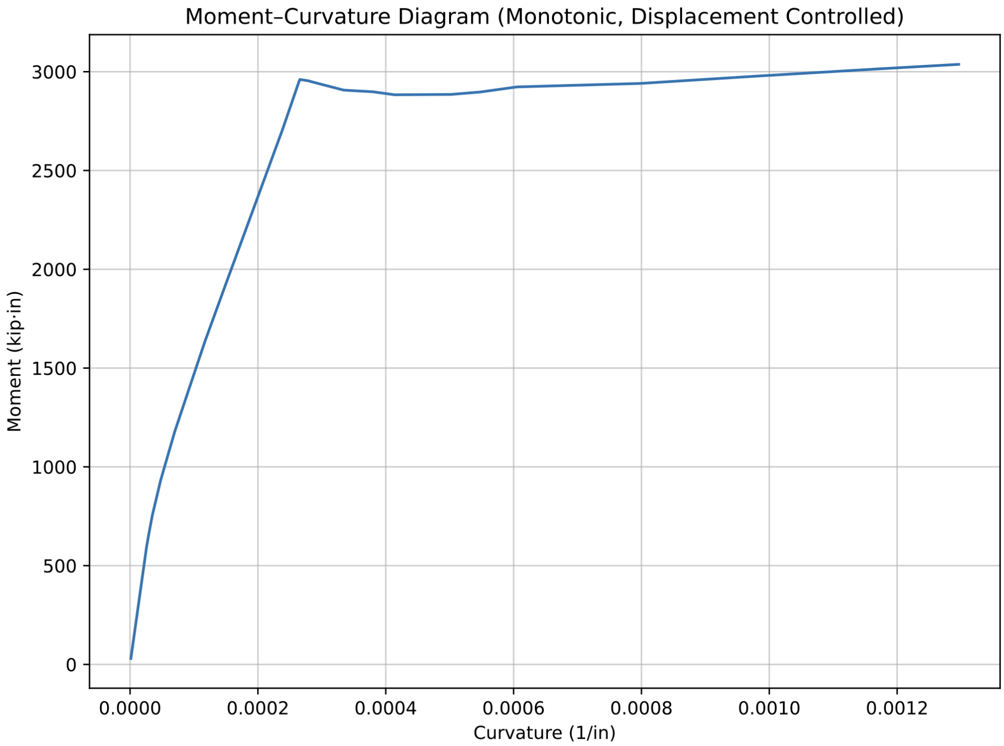


Figure 3. Moment Curvature Diagram (released Axial DOF), using Steel 01 & Concrete 02

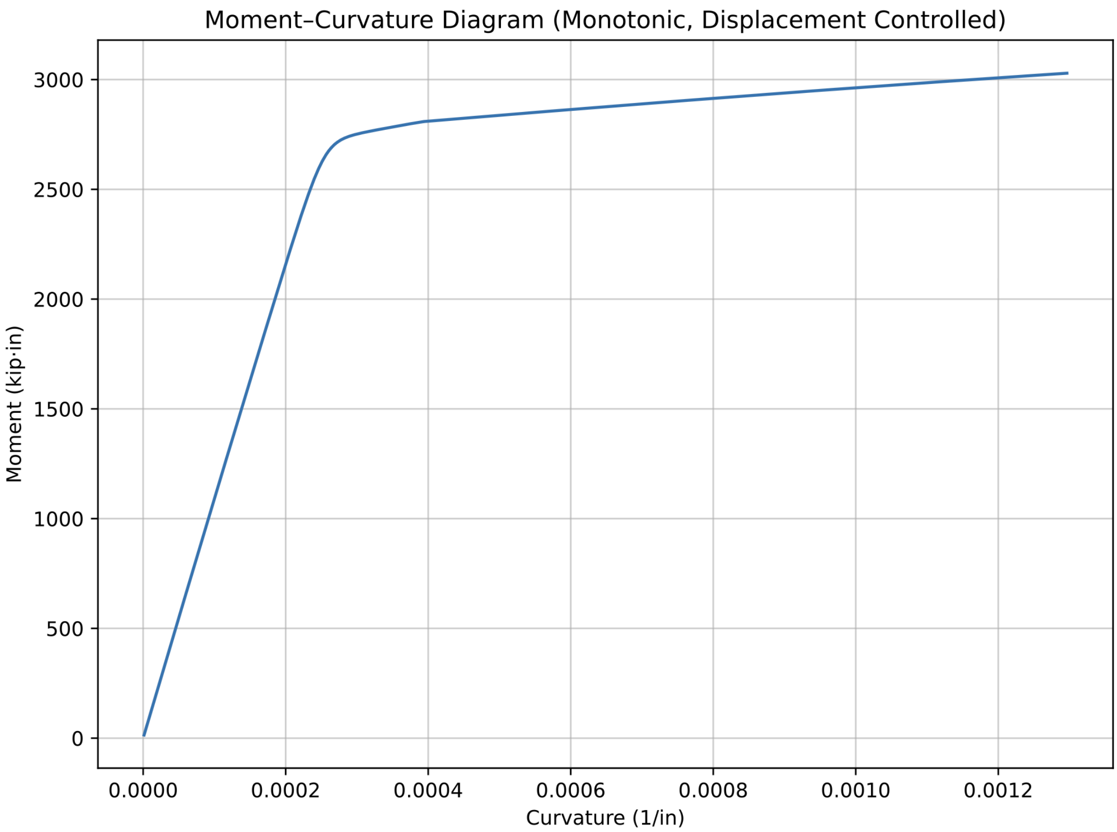


Figure 4. Moment Curvature Diagram (released Axial DOF), using Steel 02 & Concrete 01

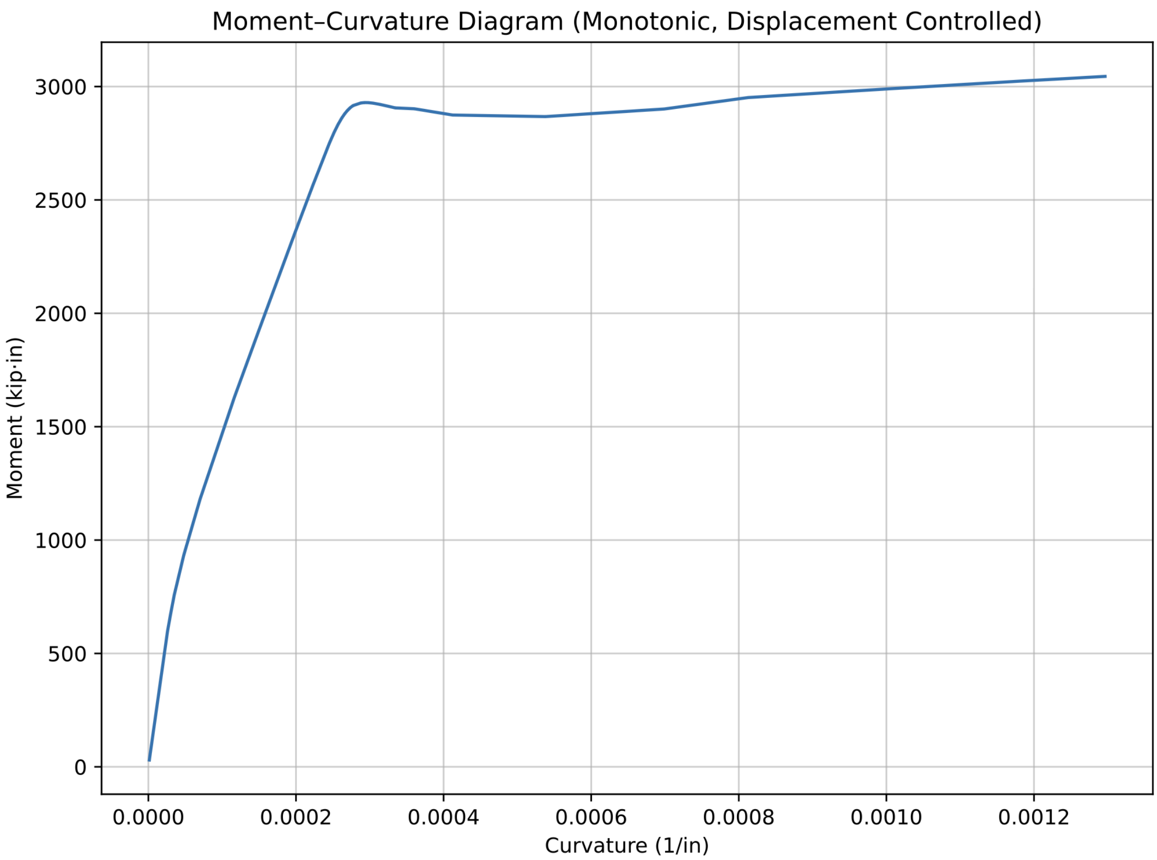


Figure 5. Moment Curvature Diagram (released Axial DOF), using Steel 02 & Concrete 02

Load control integrator

Code:

https://github.com/nirajkyadav/Homeworks/blob/main/Tutorial/Session_2_Monotonic_Load_Controlled.py

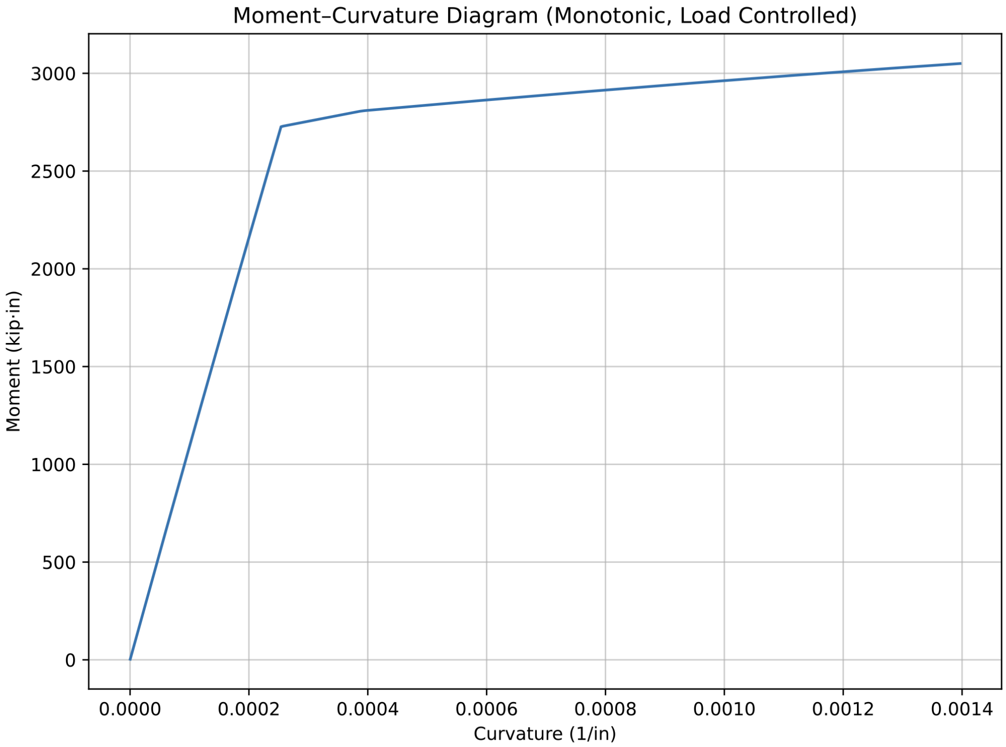


Figure 6. Moment Curvature Diagram (Load Controlled), using Steel 01 & Concrete 01

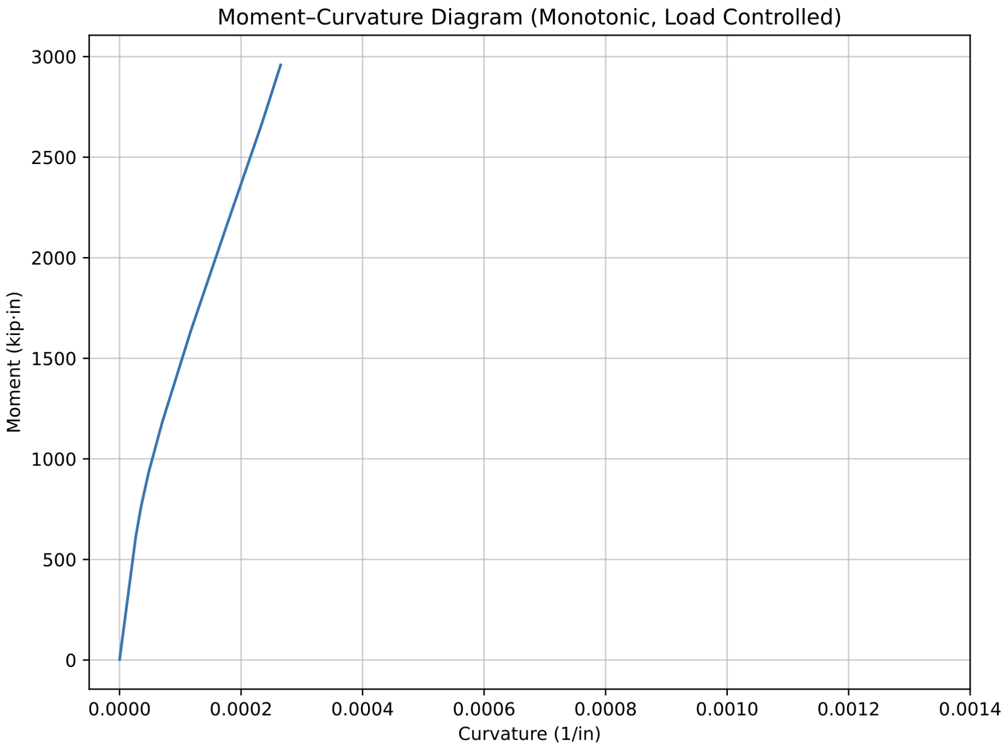


Figure 7. Moment Curvature Diagram (Load Controlled), using Steel 01 & Concrete 02

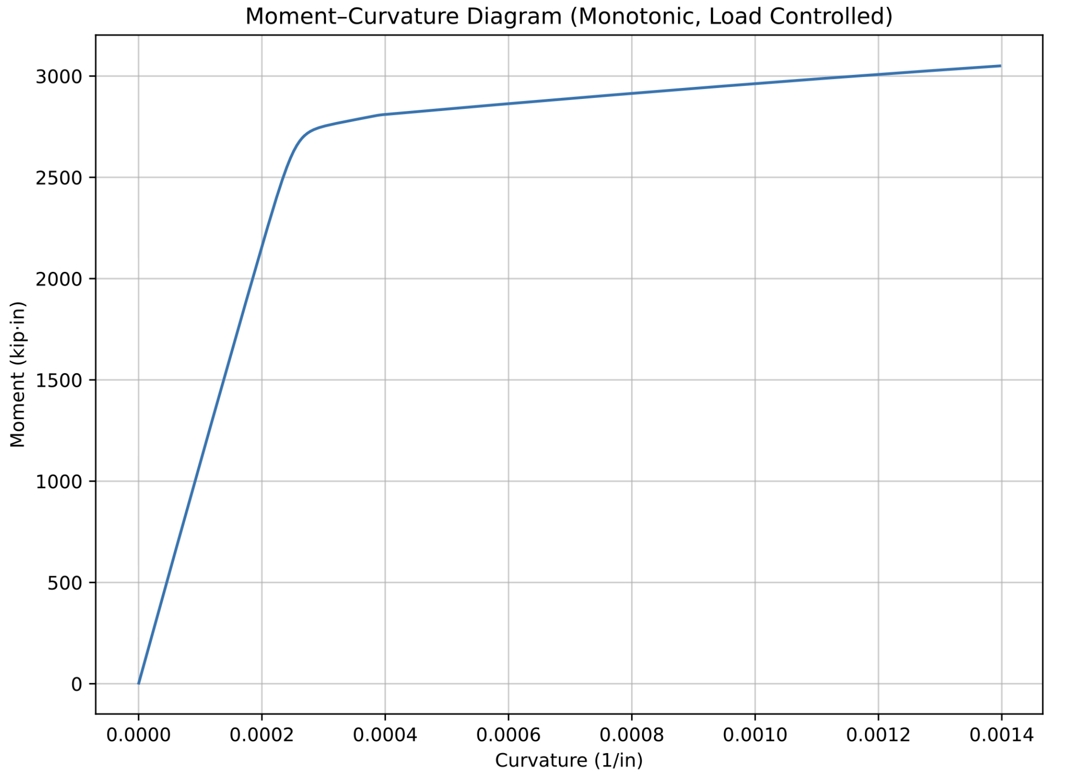


Figure 8. Moment Curvature Diagram (Load Controlled), using Steel 02 & Concrete 01

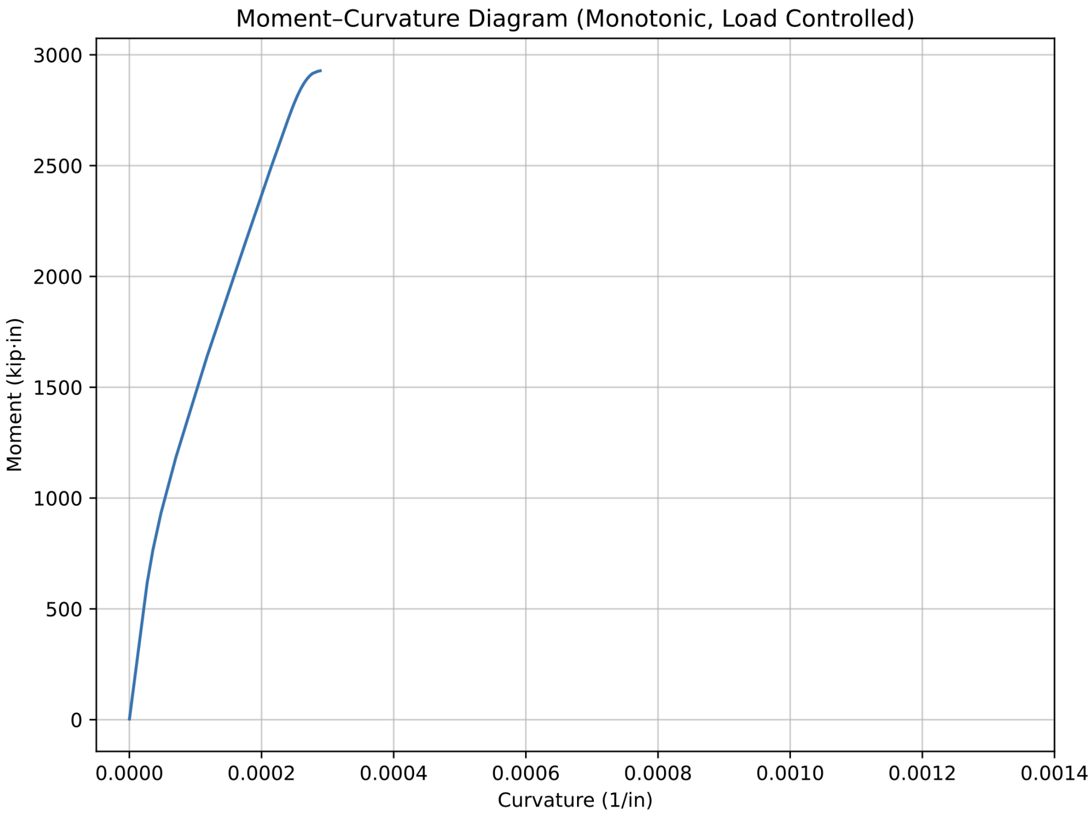


Figure 9. Moment Curvature Diagram (Load Controlled), using Steel 02 & Concrete 02

2) Cyclic Loading:

Code: https://github.com/nirajkyadav/Homeworks/blob/main/Tutorial/Session_2_Cyclic.py

Materials: Combinations of Steel02 and Concrete02
Nonlinear Analysis Algorithms:

- Newton
- NewtonLineSearch
- Broyden
- ModifiedNewton
- BFGS

Convergence Tests and Tolerances:

- RelativeNormDispIncr, Tolerance: 10^{-4}
- RelativeNormUnbalance, Tolerance: 10^{-2}
- RelativeEnergyIncr, Tolerance: 10^{-3}

The generated moment-curvature curves agree well with the expected theoretical response and accurately capture the nonlinear material behavior under cyclic loading, including the Bauschinger effect exhibited by the reinforcing steel during load reversals.

As expected for nonlinear cyclic analyses, several convergence difficulties were encountered during the simulations. I resolved them by implementing multiple fallback solution algorithms, allowing the analyses to proceed by resolving the issues.

Among the convergence criteria investigated, the RelativeNormDispIncr test with a tolerance of 10^{-4} provided the most stable and reliable performance, resulting in the highest rate of successful convergence throughout the cyclic loading history.

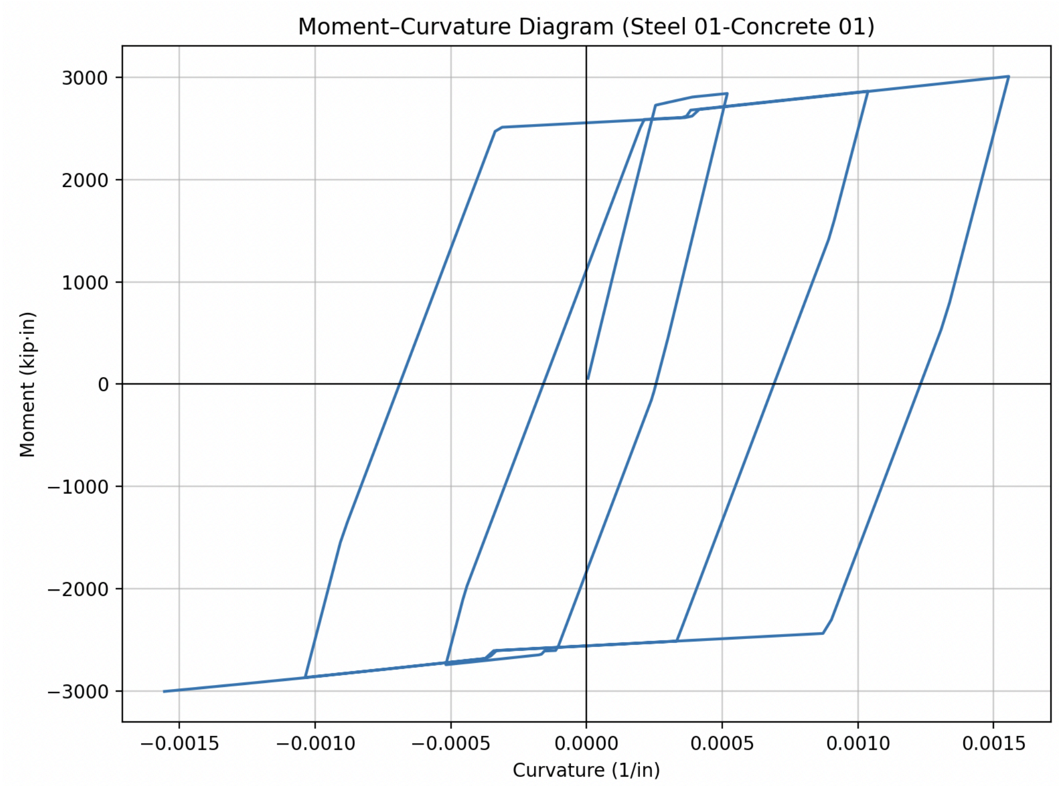


Figure 10. Moment Curvature Diagram, using Steel 01 & Concrete 01

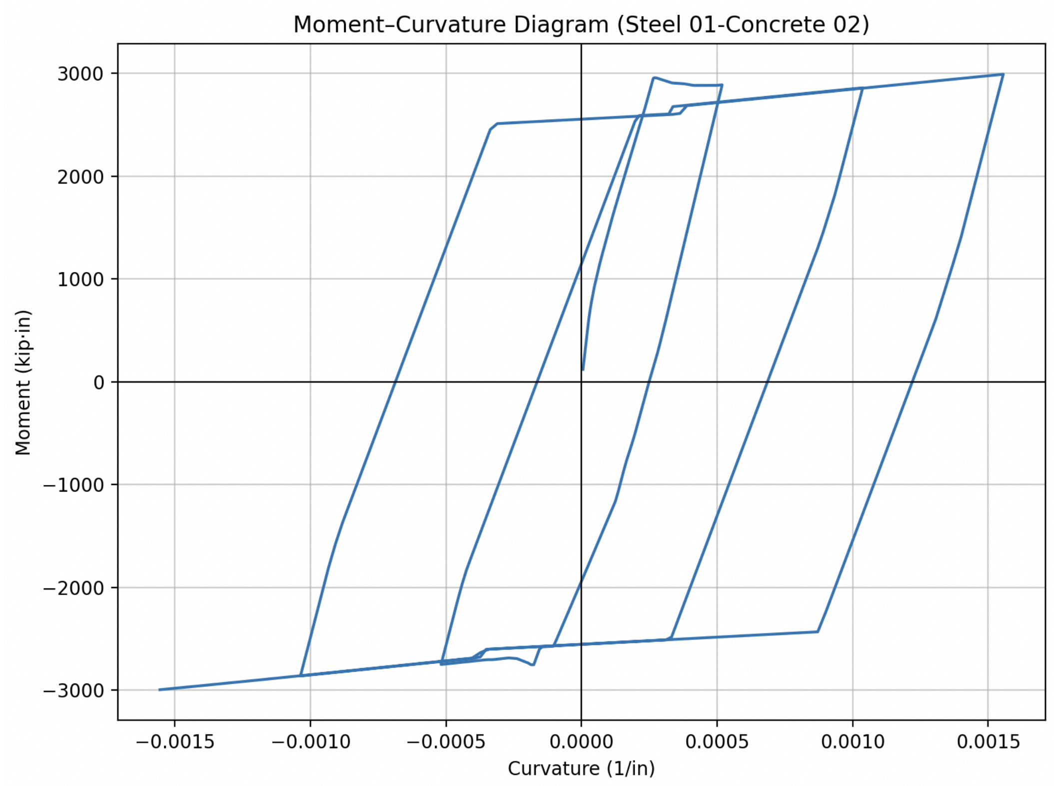


Figure 11. Moment Curvature Diagram, using Steel 01 & Concrete 02

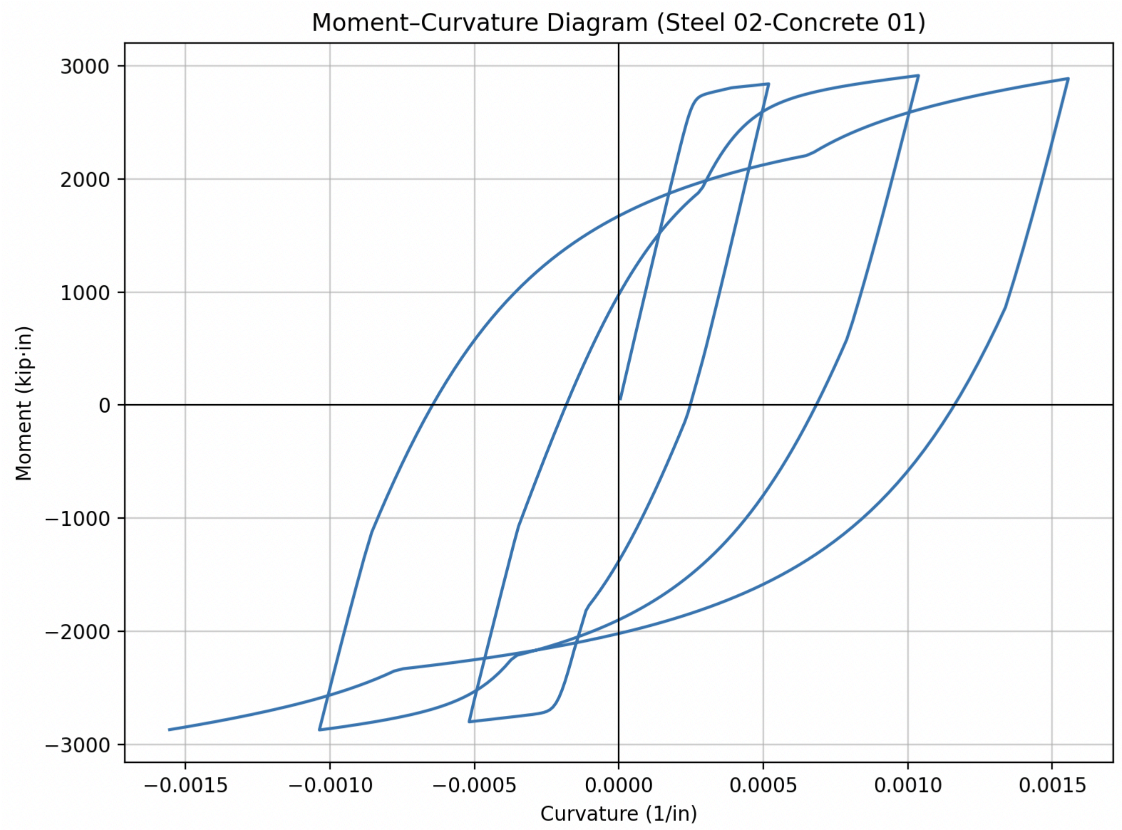


Figure 12. Moment Curvature Diagram, using Steel 02 & Concrete 01

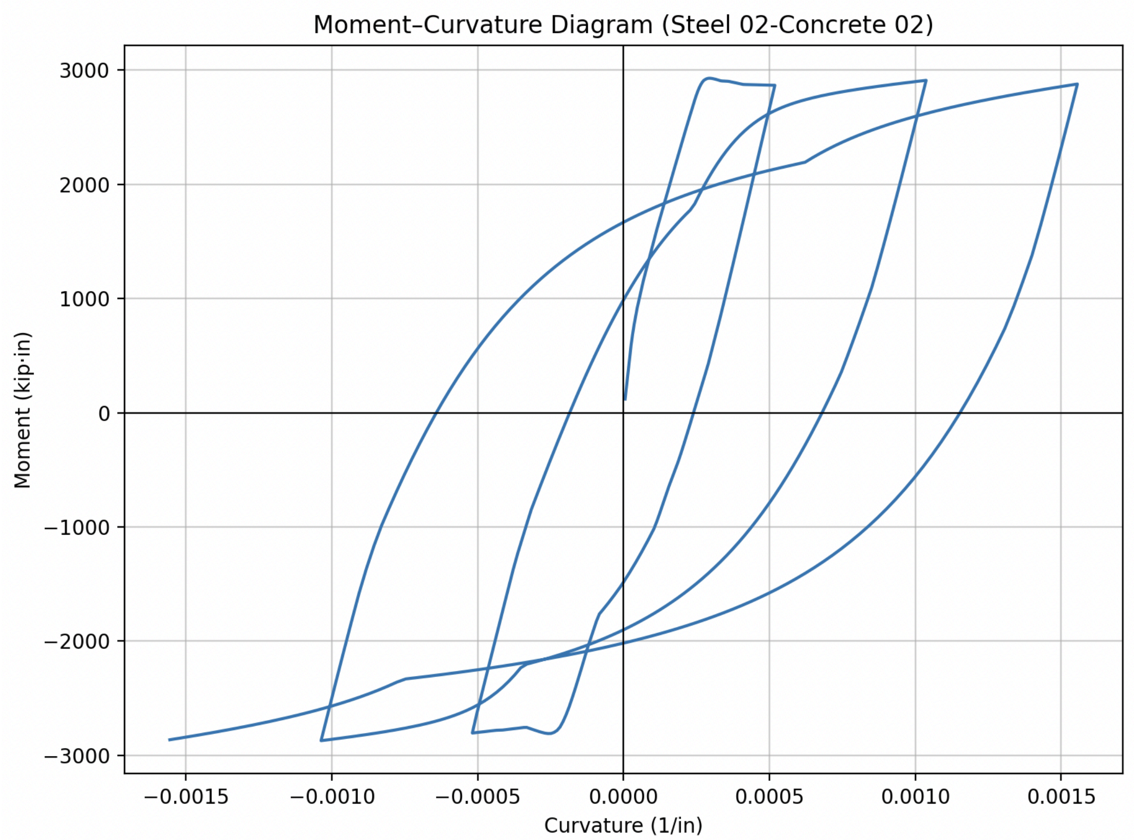


Figure 13. Moment Curvature Diagram, using Steel 02 & Concrete 02

Session 3, Quasi-Static cyclic analysis of RC cantilever Beam

Assignment: Extract Force-displacement curve using force/displacement based elements

Code:
https://github.com/nirajkyadav/Homeworks/blob/main/Tutorial/Session_3_dispBeamColumn.py
https://github.com/nirajkyadav/Homeworks/blob/main/Tutorial/Session_3_forceBeamColumn.py
https://github.com/nirajkyadav/Homeworks/blob/main/Tutorial/Session_3_Plotting.py

No of dispBeamColumn = 3
No of forceBeamColumn = 1

Displacement increment steps = 0.001

For non-convergence,
Step factors = [1.0, 0.5, 0.1, 0.05, 0.01, 0.001]

```
Convergence tests = [
   ('RelativeTotalNormDispIncr', 1.0e-4),
   ('RelativeNormDispIncr', 1.0e-4),
   ('RelativeNormUnbalance', 1.0e-2),
   ('RelativeEnergyIncr', 1.0e-3) ]

Solution algorithms = [
   ('Newton', None),              # Standard Newton-Raphson
   ('NewtonLineSearch', 0.8),
   ('Broyden', 8),
   ('BFGS', None),
   ('ModifiedNewton', None),
   ('ModifiedNewton', '-initial')
]
```

Under the prescribed cyclic loading pattern, the obtained moment-curvature curves agree well with the expected structural response and reasonably represent the nonlinear behavior of the cantilever specimen.

A significant number of convergence issues were encountered during the analysis. To address these, I reworked a robust solution framework that automatically employed alternative solution algorithms and convergence tests, while reducing the displacement increment when necessary. This strategy successfully resolved the convergence difficulties, allowing the complete response to be traced. The final converged moment-curvature curves are presented in Figure 14.

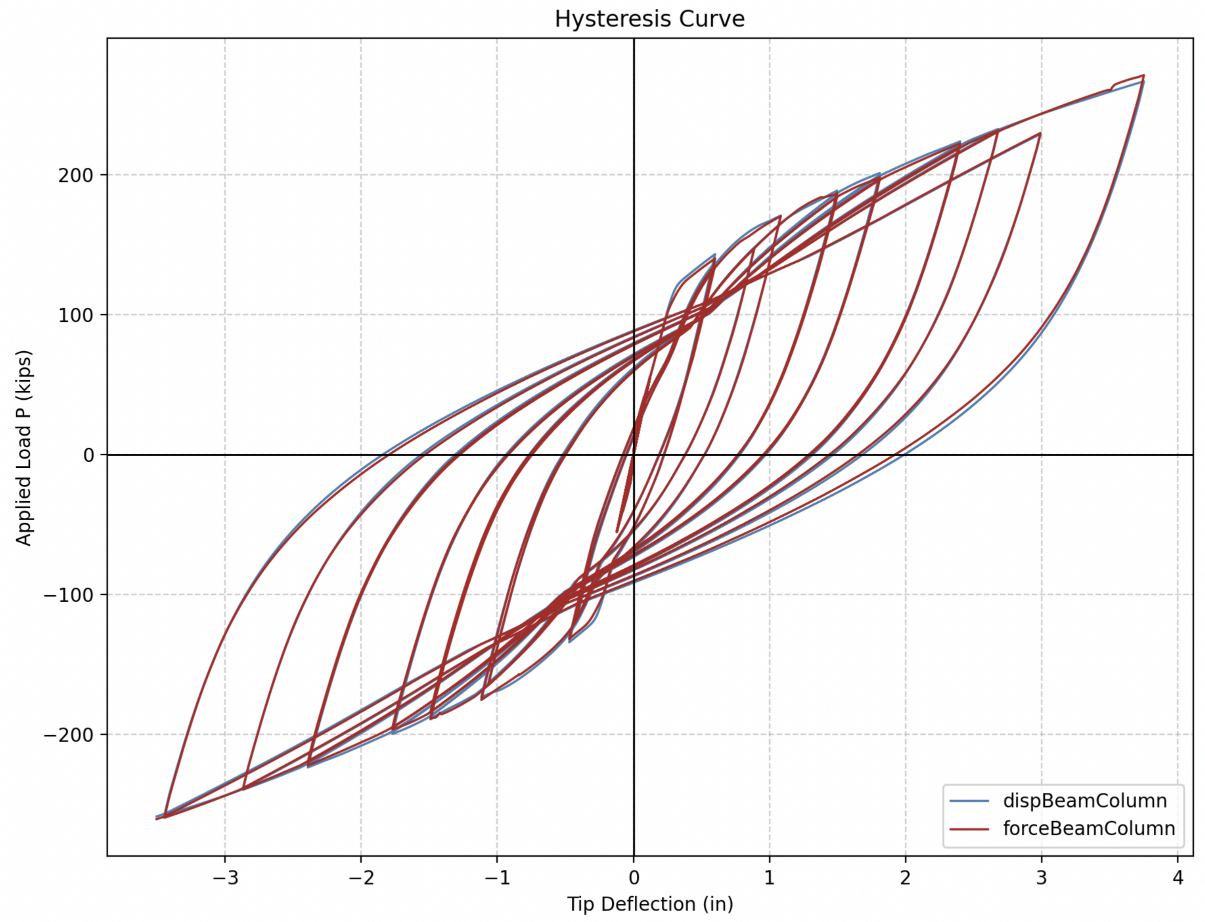


Figure 14. Hysteresis Curve Plot dispBeamColumn vs forceBeamColumn